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Python Project

Crime Exploratory Data Analysis Report

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Executive Summary

This report presents the results of an analysis of UK police street crime data across four police forces - Metropolitan, West Yorkshire, Devon & Cornwall and North Yorkshire - covering the period from January 2024 to December 2025. The analysis was carried out to help a real estate client identify areas that may be suitable for property investment based on crime patterns and trends.

The crime data was collected from publicly available sources and combined with population figures from the Office for National Statistics and median house price data. This helped provide a clearer comparison between the different regions. A number of charts and maps were created to explore crime levels, the types of crimes recorded, seasonal patterns, where crime is most concentrated geographically, and how crime levels compare with property values.

The results show a clear difference between the urban and rural police forces. Urban areas recorded higher crime levels and more concentrated crime hotspots, while the rural forces generally showed lower crime rates and more spread-out patterns. When population size was measured, West Yorkshire recorded the highest crime rate despite having the lowest house prices, suggesting a less attractive balance between risk and return for property investment. The Metropolitan Police area recorded the highest number of crimes overall but still has very strong property values and housing demand. Devon & Cornwall and North Yorkshire recorded lower crime rates and more dispersed crime patterns.

Based on these findings, the Metropolitan Police area and North Yorkshire appear to be the most suitable options for property investment. London offers strong property values and market demand, while North Yorkshire provides a lower crime environment that may appeal to investors looking for a more stable and lower-risk option.

1. Introduction

The aim of this project is to analyse UK police street crime data to identify patterns in crime across different regions. This analysis is carried out for a real estate company where crime levels can influence decisions about property investment and development. Nadine Green, Head of Sales, wants to understand which locations may be more or less desirable based on crime levels so that better decisions can be made about where the company should focus future investment.

The project explores police street crime data from four police forces: Metropolitan Police Service, Devon & Cornwall Police, North Yorkshire Police and West Yorkshire Police. The dataset covers a two-year period from January 2024 to December 2025 and includes information about individual crime incidents such as crime type, location coordinates and crime outcomes. By looking at both large urban areas such as London and more rural regions such as Devon, Cornwall and North Yorkshire, the analysis also considers how crime patterns may differ between urban and rural locations.

To support the analysis, two additional datasets were used. Population data from the Office for National Statistics was used to calculate crime rates relative to population size, and median house price data was included to provide further context when considering property investment decisions. The analysis focuses on identifying common crime types and exploring where crime is most concentrated geographically. Overall, the findings aim to provide a general overview of crime levels across the four regions and highlight areas that may require further consideration when making property-related decisions.

2. Dataset

The required dataset used in this project was the police street crime dataset, which is publicly available through the government police data website. This dataset provides monthly records of reported street crimes across different police forces in England. For this analysis, crime data from January 2024 to December 2025 was selected, as this provides recent data while still covering a long enough period to allow meaningful analysis of crime patterns.

Four police forces were chosen for the analysis: Metropolitan Police Service, West Yorkshire Police, Devon & Cornwall Police, and North Yorkshire Police. These regions were selected to provide a balance between urban and rural areas as well as representation from both the north and south of England. For example, the Metropolitan Police covers a large urban area in London, while regions such as Devon & Cornwall and North Yorkshire represent more rural locations.

In addition to the crime dataset, two additional datasets were used to support the analysis. Population data from the Office for National Statistics (ONS) was included so that crime levels could be considered relative to population size. Median house price data was also included to provide context when considering property values across the different regions. These additional datasets help provide a clearer picture when comparing crime levels across the selected police forces. Population estimates from the Office for National Statistics were taken from the 2024 release, as 2025 figures were not available at the time of the project. Median house price data was available at police force level and reflects the most recent market values at the time the analysis was carried out.

2.2 Data Pre-Processing

Before carrying out the analysis, several preprocessing steps were performed to clean and prepare the datasets. These steps ensured the data was consistent, suitable for visualisations and ready to be combined with the additional datasets used in the analysis. The main preprocessing steps included:

1. **Cleaning and testing a single dataset first**
   A single crime dataset was first loaded and cleaned to understand the structure of the data and identify potential issues such as missing values or unnecessary columns. This allowed the cleaning steps to be tested and confirmed before applying the same process to the full dataset.

2. **Applying the cleaning process to all monthly files**
   Once the cleaning process worked correctly on one dataset, Python's glob library was used to automatically load and combine all monthly crime CSV files across the selected police forces into a single dataset covering the full two-year period.

3. **Removing unnecessary columns and simplifying the dataset**
   Columns that were not useful for the analysis were removed to make the dataset easier to work with. The Context column contained no values and was removed. The Reported by and Falls within columns contained identical information and were combined into a single Police Force column. The LSOA code column was dropped and LSOA name was simplified and renamed to Area.

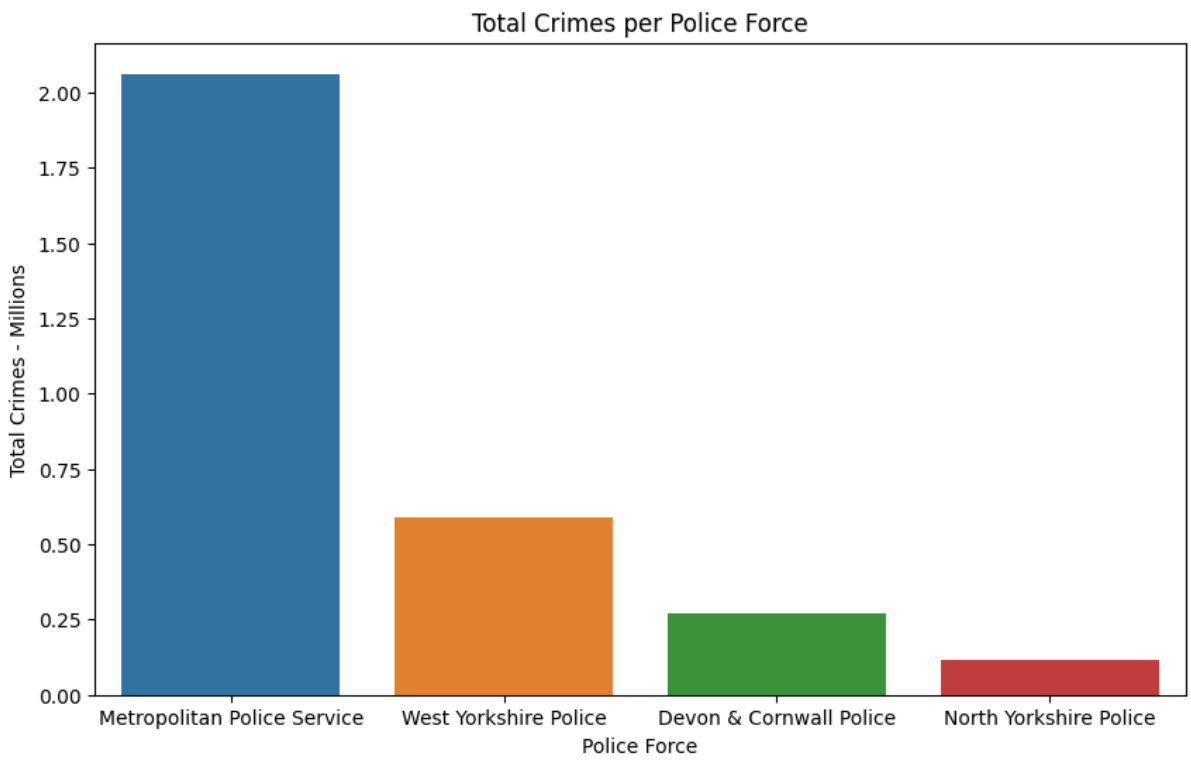
4. **Handling missing values, duplicates and validating the data**
   Rows with missing longitude or latitude values were removed since these records could not be used for geographic visualisations. Missing Crime ID and Last outcome category values were filled where appropriate, as some crime types such as anti-social behaviour do not generate these fields. Duplicate records were also removed. Scatter plots were used to validate the longitude and latitude values and confirm they fell within reasonable geographic ranges for the selected police force areas.

5. **Combining with additional datasets**
   The cleaned crime dataset was then merged with the ONS population dataset and median house price dataset. This allows the analysis to compare crime levels alongside population size and property values across the selected regions. The cleaned dataset was then exported and used in the analysis notebook.

3. Exploratory Data Analysis

This section of the report goes through the different visualisations and graphs which were created, as well as the analysis generated from them. Crime type, location, population and time are some of the factors explored in order to better understand how crime patterns vary across the selected police force areas.

3.1 Crime Distribution by Police Force



The bar chart is intended to show the total number of recorded crimes across each of the four police forces over the two-year period from January 2024 to December 2025. The Metropolitan Police Service recorded the highest crime count by a significant margin at around 2.06 million recorded crimes, followed by West Yorkshire Police at around 591,000, Devon & Cornwall at 272,000 and North Yorkshire Police recording the lowest at 114,000. Generally speaking, urban areas have higher rates of crime as there are bigger inequality gaps between individuals and a higher density of people in close proximity, which naturally will increase opportunity for crime to occur. This reflects in the fact the 2 urban police forces have the highest rates of crime.

However, these figures can be misleading as they don't account for population. Metropolitan Police Service covers the Greater London area with around 9 million people, which naturally results in higher recorded crimes compared to rural police forces such as Devon & Cornwall and North Yorkshire with lower population counts. To address this, crime rates per 100,000 people were calculated using population data from the Office for National Statistics in order to allow a more balanced comparison between the regions.

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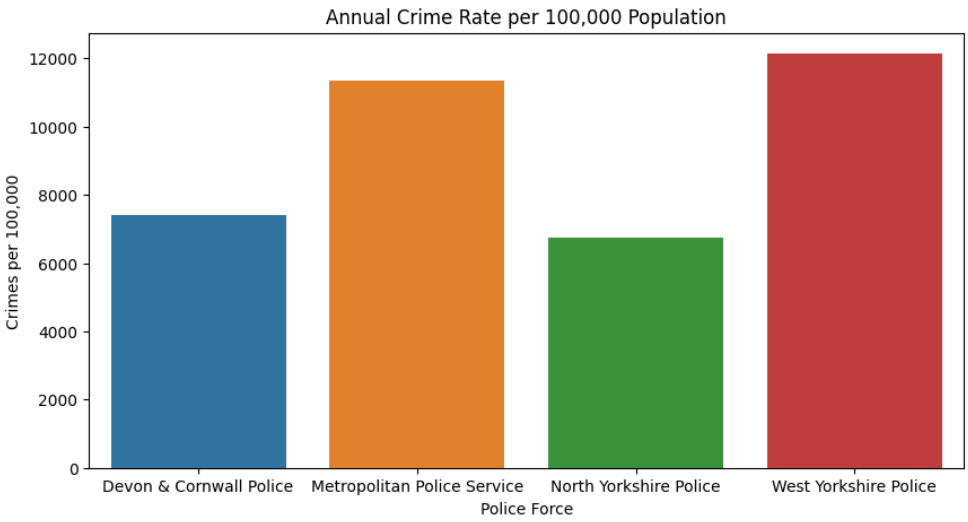
3.2 Crime Rate per 100,000 Population

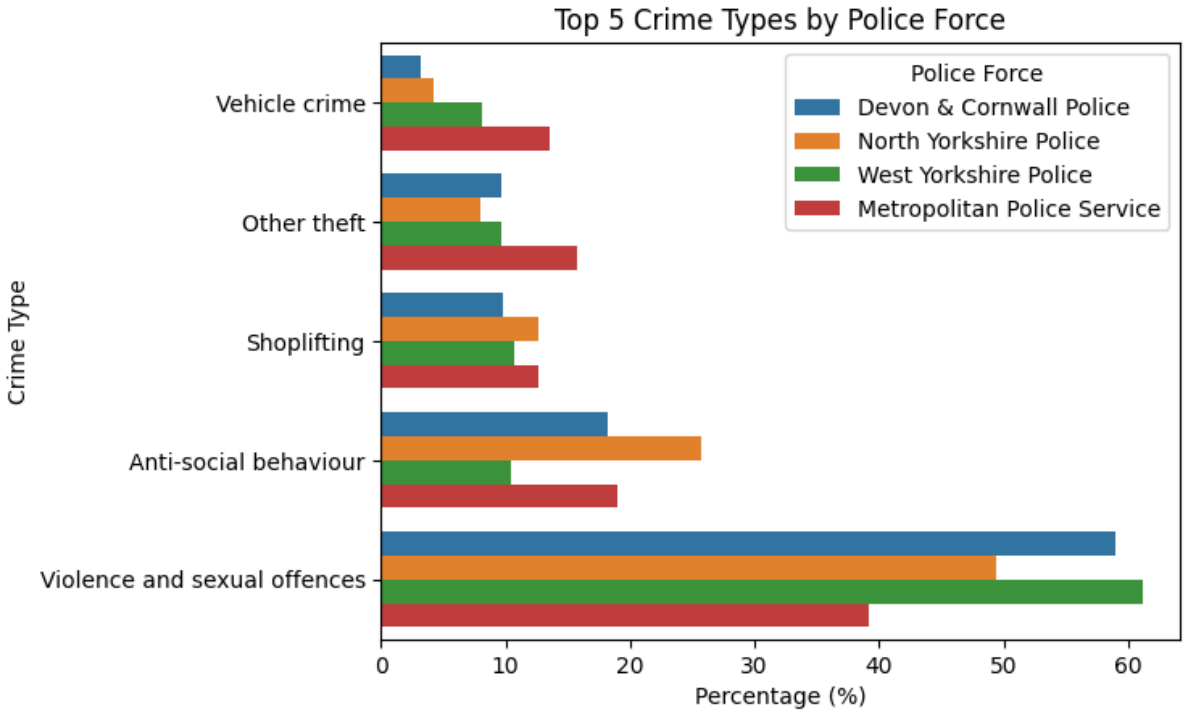


	Police_Force	Population_2024
0	Metropolitan Police Service	9,074,625
1	North Yorkshire Police	844,571
2	West Yorkshire Police	2,435,236
3	Devon & Cornwall Police	1,840,161

When accounting for the population size, the crime rate significantly changes compared to the raw crime count. The bar chart shows the annual crime rate per 100,000 population for each of the 4 police forces. West Yorkshire actually records a higher crime rate at around 12,000 per 100,000 people, narrowly ahead of the Metropolitan Police Service at around 11,300. The 2 rural areas record the lowest crime rates, however relative to their population they are significantly higher than expected from the initial crime count. Despite the Metropolitan Police Service having over four times the population of West Yorkshire, it records a lower crime rate per 100,000 suggesting that London's population size was inflating its crime figures rather than reflecting a genuinely higher risk environment.

This is a significant finding as it shows that when population is accounted for police forces such as Metropolitan and West Yorkshire are more comparable. The graph also suggests that North Yorkshire and Devon & Cornwall are genuinely lower crime areas rather than appearing lower due to smaller populations, making them more attractive regions when assessing relative safety across the four forces.

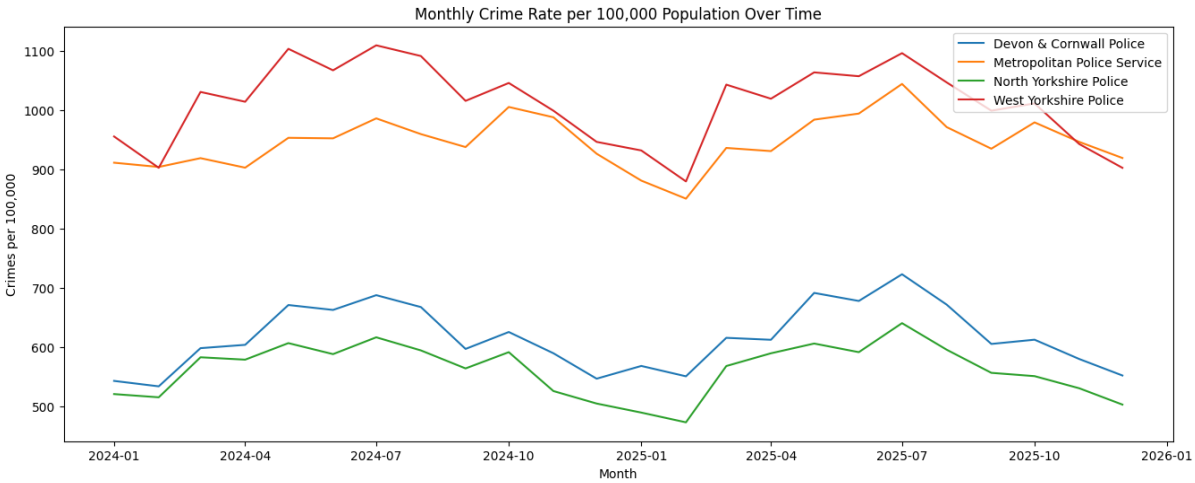
3.3 Crime Type Distribution



The bar chart shows the top 5 crime types as a percentage of total crimes for each of the four police forces. Violence and sexual offences is the most common crime type across all four forces, accounting for around 60% of crimes in both Devon & Cornwall and West Yorkshire, around 50% in North Yorkshire and 40% in the Metropolitan Police Service. The pattern across both urban and rural forces suggests that violence and sexual offences is a nationwide pattern rather than being a problem in a particular region.

However, there are some differences between forces. The Metropolitan Police record a noticeably higher proportion of other theft and vehicle crime compared to the other three forces, which is typical of a dense urban environment where there is greater opportunity for opportunistic crime. North Yorkshire records the highest proportion of anti-social behaviour at around 26%, which may reflect the difficulty of policing a large rural area where community disturbances are more commonly recorded in the absence of other crime types. Overall, the broad pattern across all four of the forces suggest that the type of crime an area experiences Is less dependant on whether its urban or rural and more reflective national trends.

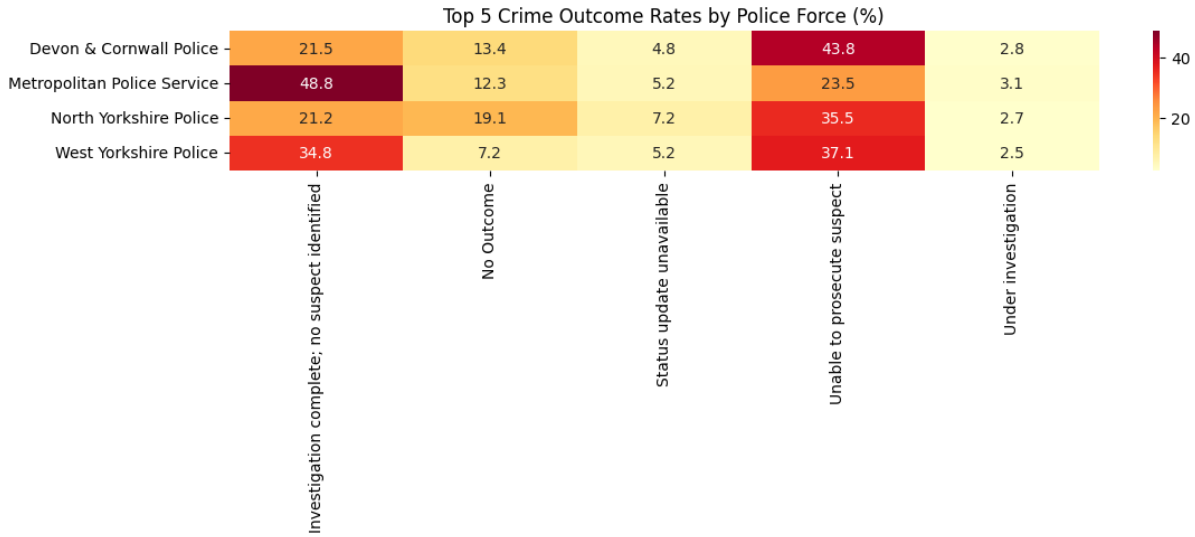
3.4 Crime Trends Over Time



The line graph shows the monthly crime rate per 100,000 people for each of the four police forces between January 2024 and December 2025. A clear difference can be seen between the urban and more rural areas across the whole period. West Yorkshire and the Metropolitan Police generally record the highest crime rates, usually above 900 crimes per 100,000 people, while Devon & Cornwall and North Yorkshire remain noticeably lower, mostly between 500 and 700.

A seasonal pattern can also be seen across all four forces. Crime rates tend to increase during the summer months and fall during the winter. This may be linked to people spending more time outdoors and increased activity during warmer months, which can create more opportunities for crime to occur. West Yorkshire records the highest rates overall, reaching around 1,100 crimes per 100,000 people in mid-2024, while North Yorkshire consistently records the lowest rates across the two-year period. The gap between the urban and rural police forces stays fairly consistent throughout the timeframe, suggesting that crime levels across these regions have remained relatively stable during the period analysed.

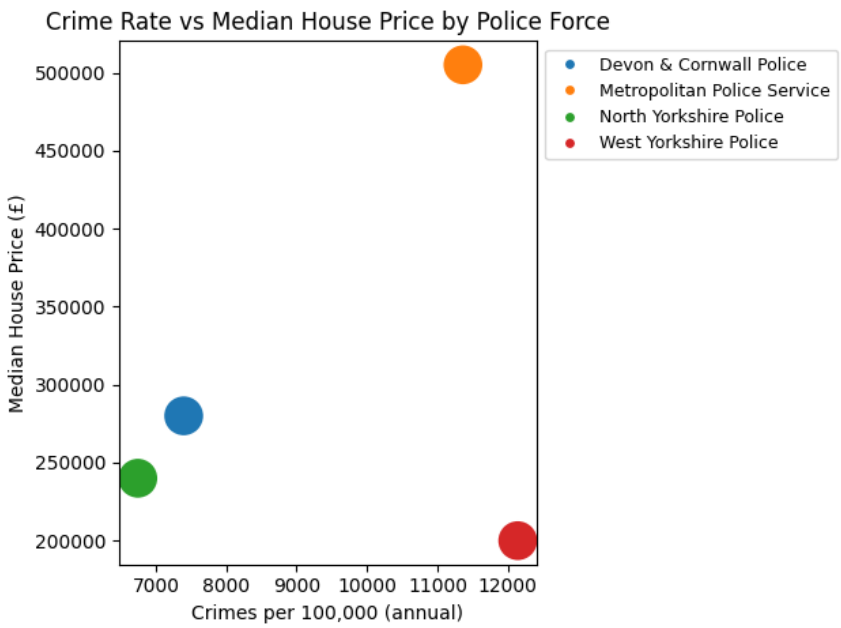
3.5 Crime Outcomes by Police Force



The heatmap shows the top five crime outcomes as a percentage of total recorded crimes for each of the four police forces. These outcomes were chosen because they make up the majority of all recorded outcomes across the dataset. One of the clearest patterns is that a large proportion of crimes across all four forces end with either no suspect being identified or the suspect being unable to be prosecuted. This suggests that many crimes remain unresolved regardless of whether the area is more urban or rural.

The Metropolitan Police have the highest rate of investigations being completed with no suspect identified, at around 48.8%. This indicates that although it is the largest police force, a significant share of crimes still remains unsolved. Devon & Cornwall records the highest rate of suspects being unable to be prosecuted at 43.8%, while North Yorkshire shows the highest proportion of cases recorded as having no outcome at 19.1%. This may be related to the higher number of anti-social behaviour incidents seen earlier in the analysis, as these cases often do not result in any formal outcome. Overall, the heatmap suggests that crime resolution rates are relatively low across all four police forces, which is an important factor to consider when comparing safety and policing effectiveness across the regions.

3.6 Crime Rate vs Median House Price



	Police Force	Median House Price
0	Devon & Cornwall Police	280000
1	Metropolitan Police Service	505000
2	North Yorkshire Police	240000
3	West Yorkshire Police	200000

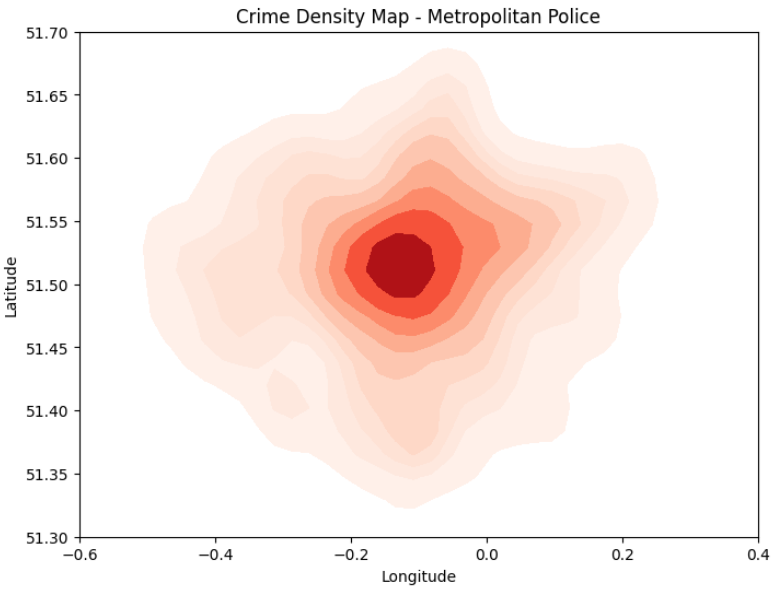
This scatter plot compares the annual crime rate per 100,000 people with the median house price across the four police force areas. The Metropolitan Police area appears in the upper right of the chart, with both a high crime rate of around 11,300 per 100,000 people and the highest median house price at roughly £505,000. This shows that higher crime does not always lead to lower house prices. In London's case, property demand is influenced by many other factors such as employment opportunities, transport links and the overall economic activity of the city, which help maintain high property values despite higher crime levels.

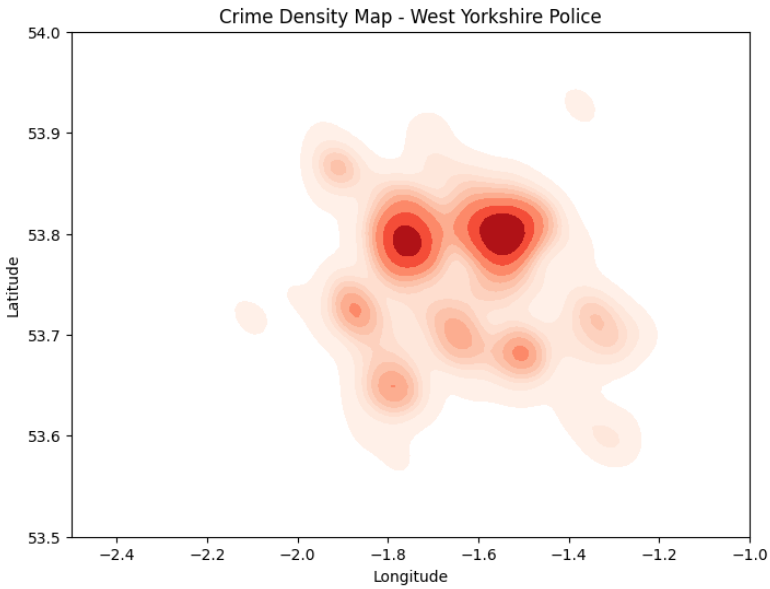
West Yorkshire stands out in a different way. It records the highest crime rate at around 12,000 per 100,000 people while also having the lowest median house price at around £200,000. This creates a less attractive balance between risk and return, as higher crime combined with lower property values may reduce the potential appeal for property investment. In comparison, London's high crime rate is balanced by much higher property prices and strong housing demand, meaning the potential returns may still justify the risk. The two more rural forces are on the lower left of the chart, showing lower crime rates alongside mid-range house prices. Devon & Cornwall has a median house price of around £280,000 and North Yorkshire around £240,000. This suggests that rural areas may offer a combination of lower crime and more affordable property prices, which could represent a more stable and lower-risk investment option.

3.6 Geographical Distribution of Crime

The following maps use kernel density estimation to show where crime is most concentrated geographically within each of the four police force areas. The forces are grouped into urban and rural pairs so that it is easier to compare how crime is distributed across different types of regions.

3.6.1 Urban Forces – Metropolitan & West Yorkshire Police Force

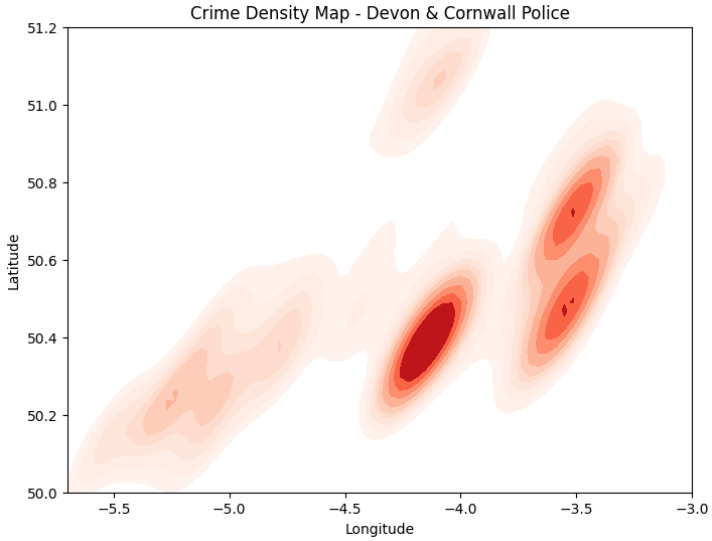


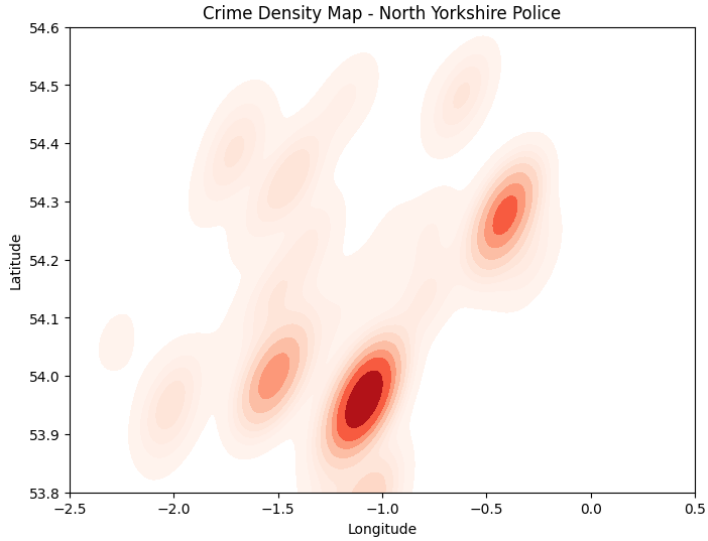


Both urban police forces show clear crime hotspots, although the patterns are slightly different between the two. In the Metropolitan Police area, crime is heavily concentrated in central London, particularly in areas such as Westminster, Camden and Islington. These locations attract large numbers of tourists, commuters and nightlife activity, which naturally increases the chances of crime occurring. This pattern also reflects what was seen earlier in the crime rate analysis, where London showed a high overall crime rate that is largely driven by its busy and densely populated central areas. Moving further away from the centre, crime density gradually becomes lower. Outer boroughs such as Bromley or Richmond appear to have fewer concentrated crime areas, suggesting they may offer a lower crime environment while still benefiting from London's strong property market and demand.

West Yorkshire shows a different pattern. Instead of one main centre of crime, there are two major hotspots driven by the cities of Leeds and Bradford. Both cities have their own concentrated crime areas, with smaller hotspots also appearing around surrounding towns such as Huddersfield and Wakefield. This creates a more spread-out pattern of crime across the region rather than one single area. As a result, crime is more widely distributed across the police force area. When considered alongside the house price analysis earlier, this reiterates the less favourable risk-to-return ratio for West Yorkshire.

3.6.2 Rural Forces – Devon & Cornwall & North Yorkshire Police Force





In comparison to the urban police forces, Devon & Cornwall and North Yorkshire show a very different pattern on the maps. Instead of large, concentrated areas of crime, incidents appear in smaller clusters with large areas in between where little or no crime is recorded. This reflects the more spread-out nature of rural populations, where towns and communities are separated by large distances.

In Devon & Cornwall, crime tends to appear around the main towns and coastal population centres. While there are a few areas where crime is more concentrated, these hotspots are much smaller than those seen in the urban police forces and more separate from each other. This suggests that even in the busiest parts of the region, crime remains fairly localised.

North Yorkshire shows the most spread-out pattern of all four police forces. The hotspots are small and clearly separated from each other, with large areas of the county showing very little crime. This reflects the largely rural nature of the region, where crime tends to occur in specific towns rather than being spread across the wider area. Compared with the urban maps, the difference is very clear. While London and West Yorkshire show large areas where crime is concentrated, North Yorkshire shows crime occurring in smaller pockets surrounded by areas with very low levels of crime. This matches the lower crime rates seen earlier in the analysis and supports the idea that the rural police forces generally experience lower levels of crime overall.

4. Conclusion

This analysis looked at street crime data across four police forces in England between January 2024 and December 2025. To conclude, the results showed a clear difference between the urban and rural areas in both how much crime occurs and how it is spread across each region. At first, the Metropolitan Police appeared to have the most crime because it recorded the highest number of incidents. However, once population was considered, West Yorkshire actually had the highest crime rate. This shows that looking at total crime numbers alone can give a misleading picture when areas have very different population sizes.

The types of crimes recorded were fairly similar across all four police forces. This suggests that crime patterns are influenced more by wider trends rather than the specific region itself. A seasonal pattern could also be seen in the data, with crime levels generally increasing during the summer months and dropping slightly during winter. Another clear finding was that many crimes do not lead to a clear outcome, with a large number of cases recorded as having no suspect identified.

The maps also showed clear differences between the urban and rural areas. In the urban police forces, crime was heavily concentrated in city centres and built-up areas. In contrast, the rural forces showed smaller clusters of crime with large areas where very little crime was recorded. When this is considered alongside house prices, it shows that the link between crime and property values is not always simple. Some places with higher crime still have very expensive housing, especially in major cities where other factors such as jobs, transport and demand for housing play a big role.

Overall, the findings show clear differences between the urban and rural police forces in both crime levels and how crime is distributed across each region. The analysis also highlights that population size should be considered when comparing areas, as raw crime counts alone can give inaccurate readings.

5. Limitations

Some limitations include:

- Population data was only available for 2024, so the same figures were used for 2025 crime rate calculations.
- House price data was only available at police force level, not for smaller areas.
- The analysis only included four police forces, so it does not represent the whole of England.
- Due to time constraints, some parts of the analysis could be improved or explored further. This left parts of the project to rely on external sources i.e. creating density maps as it was outside of learning scope.

6. Recommendations

Based on the findings of the analysis, two police force areas stand out as the most suitable options for property investment: the Metropolitan Police area and North Yorkshire.

Although the Metropolitan Police area records relatively high crime rates, it still represents a strong investment option due to the much higher property values and continued demand for housing in London. The geographic maps showed that crime is mainly concentrated in central areas such as Westminster and Camden. This suggests that focusing investment in outer London boroughs, where crime levels are lower, could allow investors to benefit from the strength of the London property market while avoiding the areas where crime is most concentrated.

North Yorkshire represents the lower risk option. It recorded the lowest crime rate per 100,000 people out of the four police forces analysed. The density maps also showed that crime is spread across small, isolated areas rather than large area zones. With moderate house prices and lower crime levels, North Yorkshire could offer a more stable environment for property investment, particularly for those looking outside of major urban areas.

References

- These are the websites used to generate the median house prices.

https://www.plumplot.co.uk/London-house-prices.html

https://www.plumplot.co.uk/West-Yorkshire-house-prices.html

https://www.plumplot.co.uk/Devon-house-prices.html

https://www.plumplot.co.uk/Cornwall-house-prices.html

https://www.plumplot.co.uk/North-Yorkshire-house-prices.html

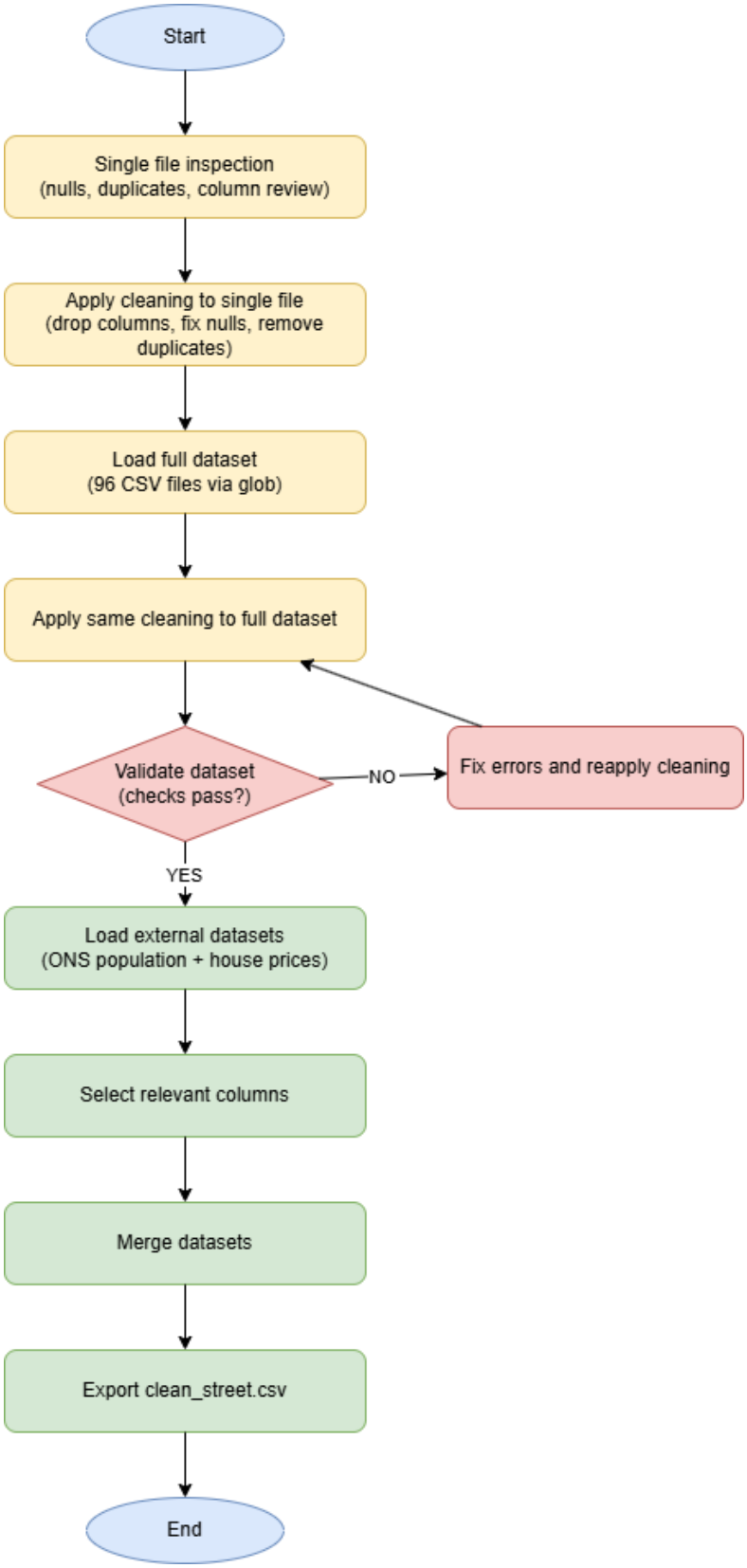
- Open street map link used for co-ordinates

https://www.openstreetmap.org/#map=5/54.91/-3.43

- Population link.

https://www.ons.gov.uk/peoplepopulationandcommunity/populationandmigration/populationestimates/adhocs/3194populationestimatesforpoliceforceareasinenglandandwalesbysingleyearofageandsexmid1991tomid2024

Appendix


Start
Single file inspection
(nulls, duplicates, column review)
Apply cleaning to single file
(drop columns, fix nulls, remove duplicates)
Load full dataset
(96 CSV files via glob)
Apply same cleaning to full dataset
Validate dataset
(checks pass?)
NO
Fix errors and reapply cleaning
YES
Load external datasets
(ONS population + house prices)
Select relevant columns
Merge datasets
Export clean_street.csv
End